

OpenSource CDS | User Guide

Documentation on functionality and calibration

Project name: BEP PME-A06

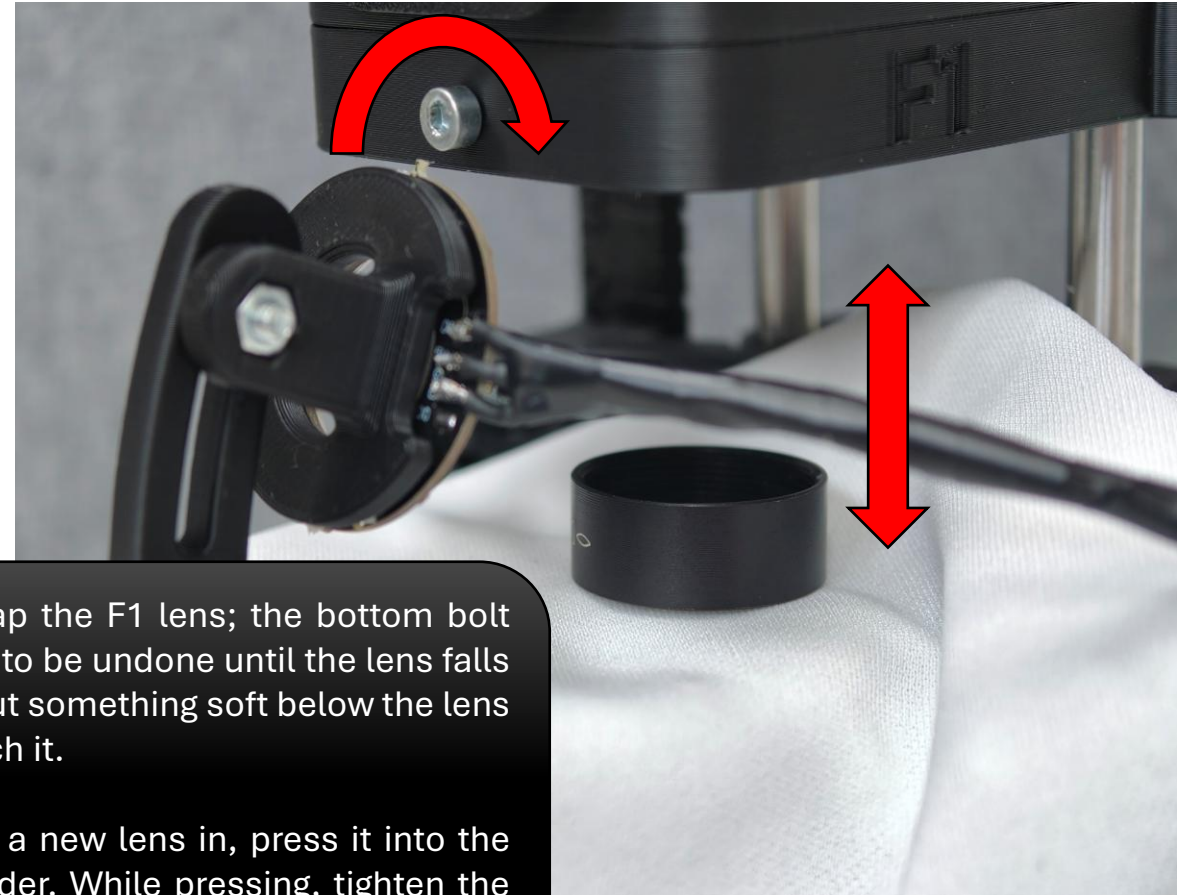
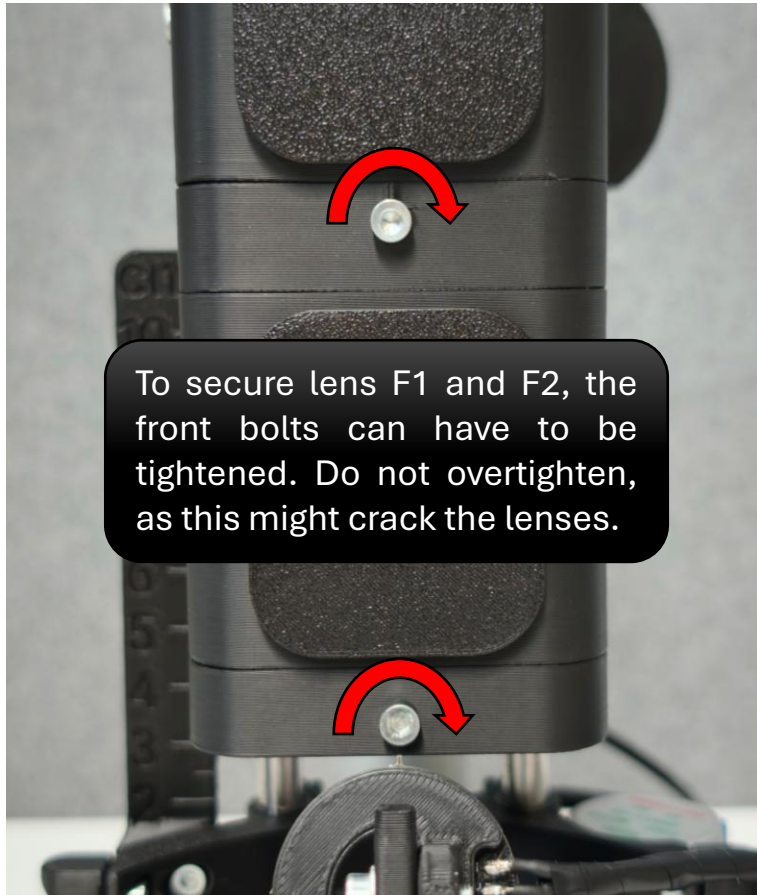
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Securing and swapping lenses/detector



Securing and swapping lenses/detector

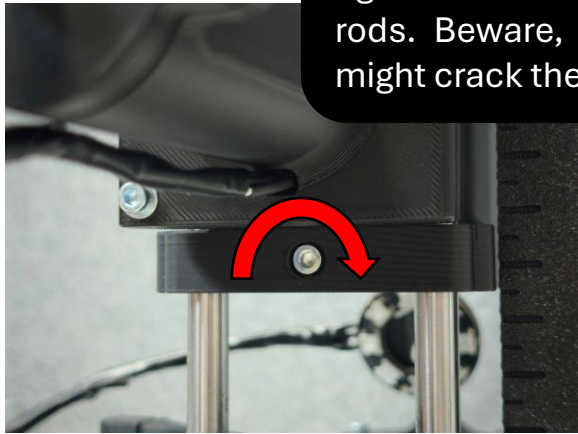


To secure the detector, slide the ring on the detector into the holder. Then, tighten the top bolt.

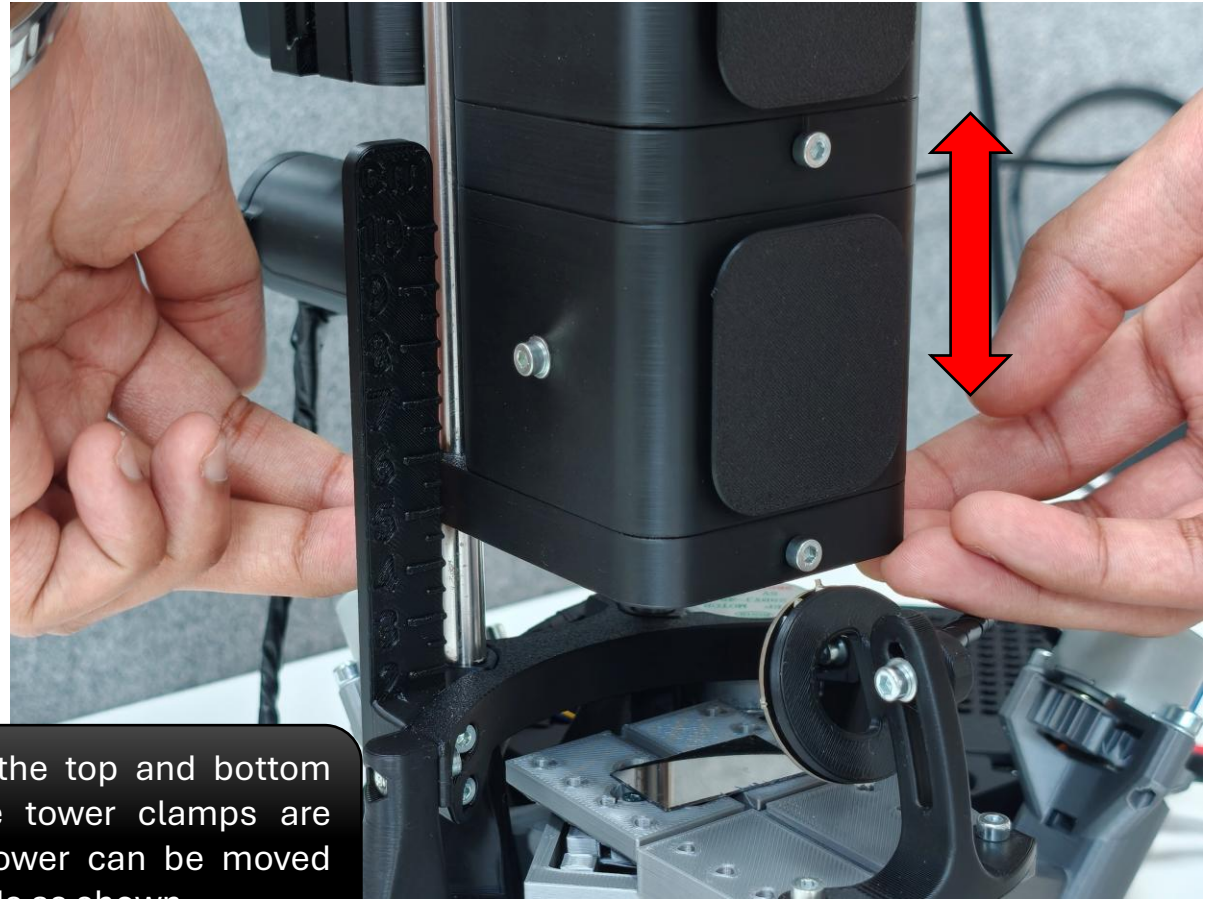
Adjustability | Tower position



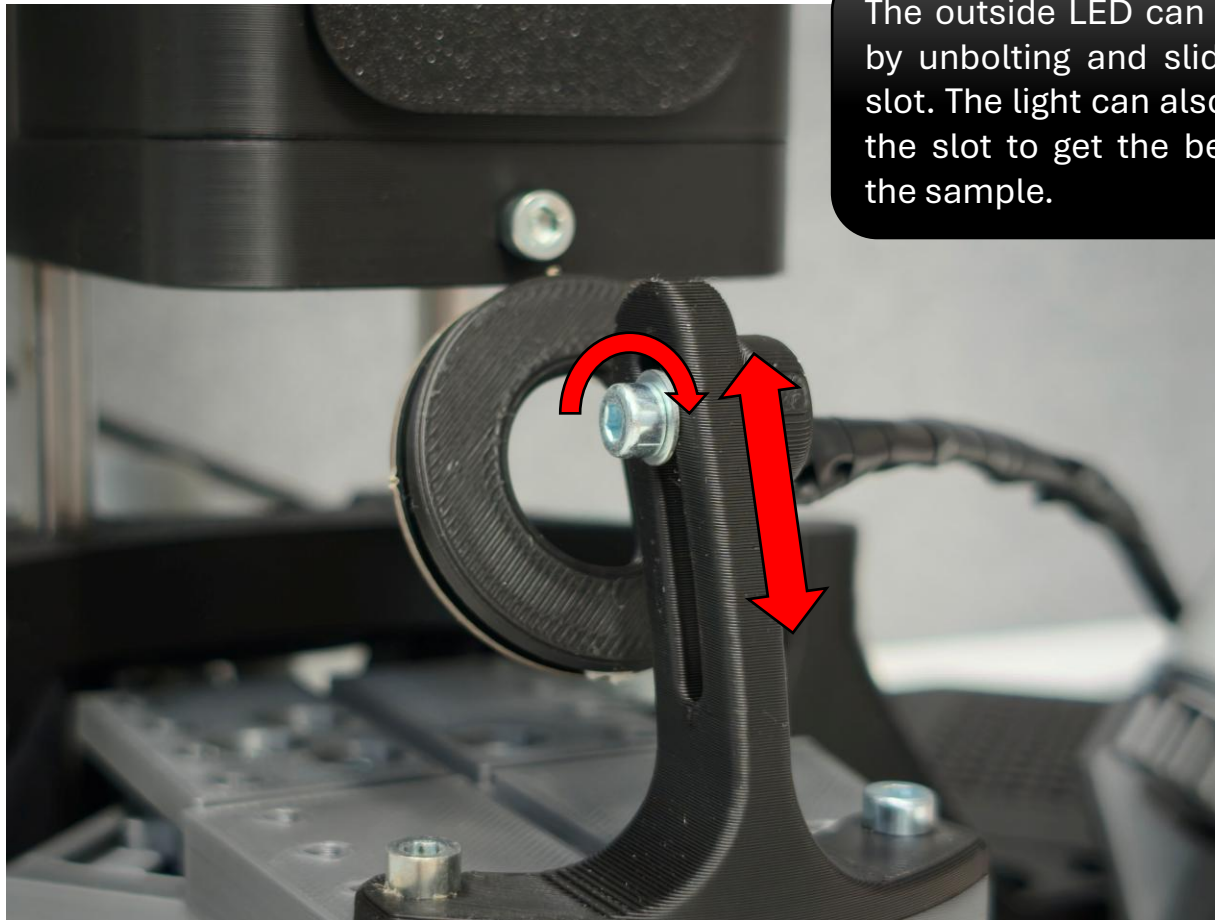
To secure the tower, the bolt at the top and bottom can be tightened to clamp onto the rods. Beware, overtightening might crack the part!



When both the top and bottom bolts of the tower clamps are loose, the tower can be moved along the rods as shown.

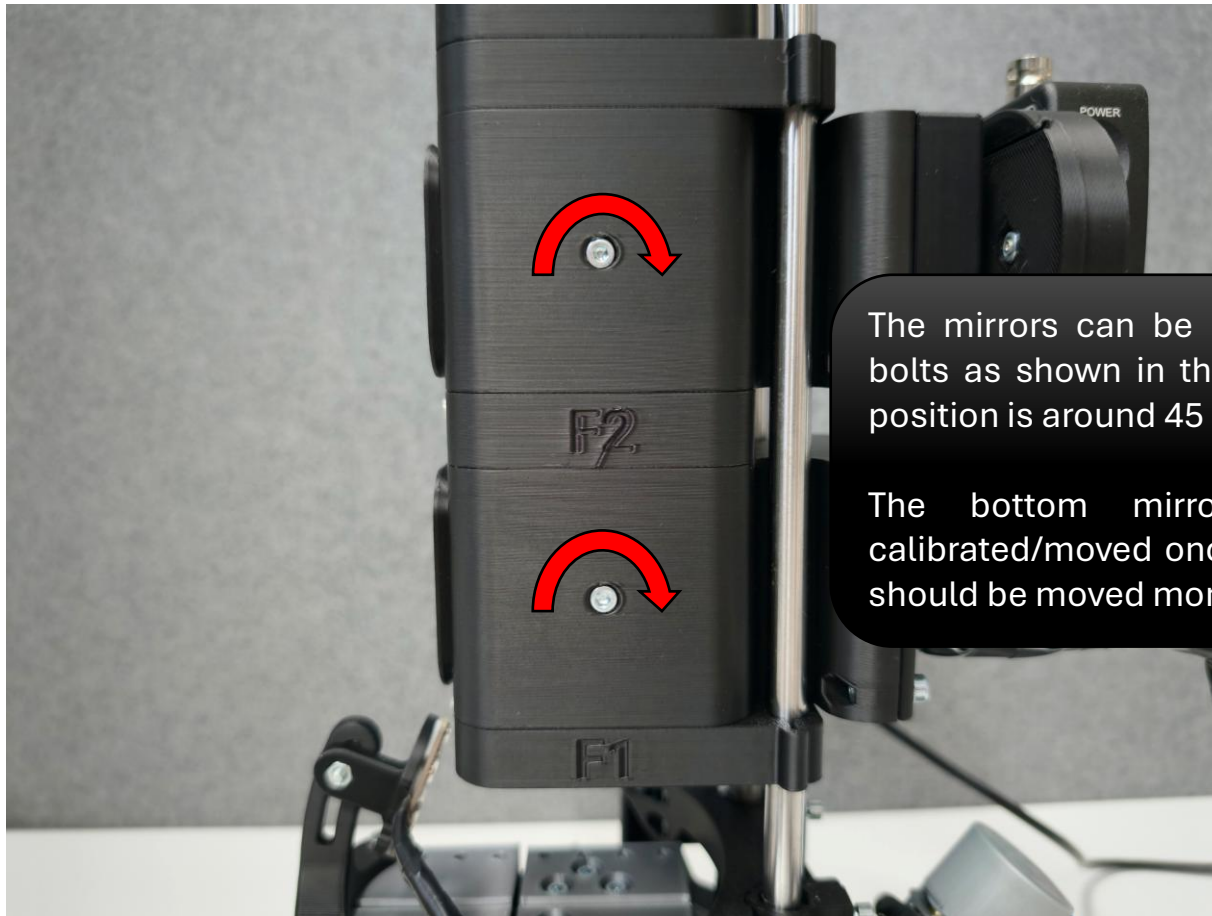


Adjustability | Outside LED position



The outside LED can be positioned by unbolting and sliding along the slot. The light can also be rotated in the slot to get the best lighting on the sample.

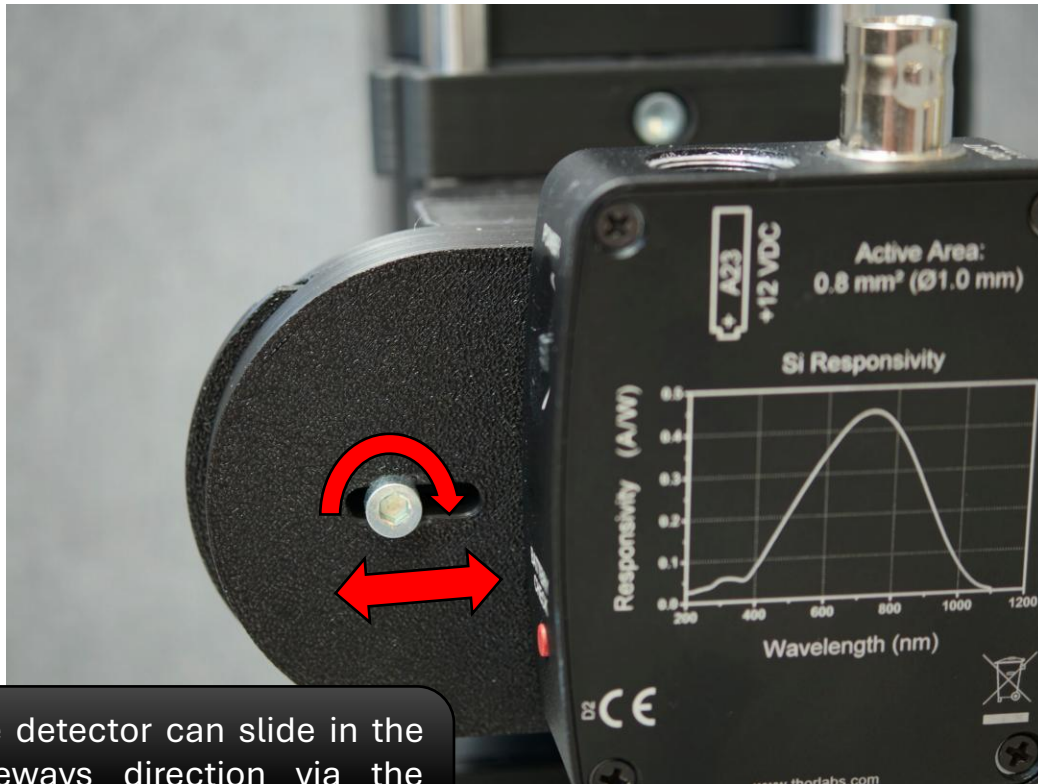
Adjustability | Mirror position



The mirrors can be rotated by turning the bolts as shown in the figure. The preferred position is around 45 degrees.

The bottom mirror should only be calibrated/moved once. Only the top mirror should be moved more than once.

Adjustability | Detector position

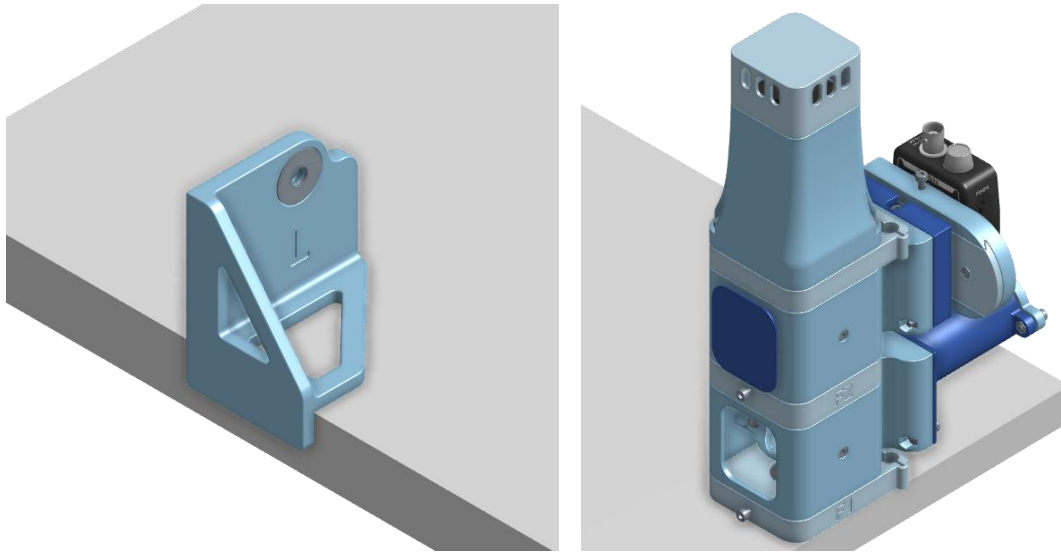


The detector can slide in the sideways direction via the slot. This is used to align the laser with the detector spot.



The detector can also slide in the 'q' direction. This is used to calibrate the offset, which puts the detector in the most sensitive part of the curve. For ease of install, the offset is best when the bolt is exactly in the middle of the slot.

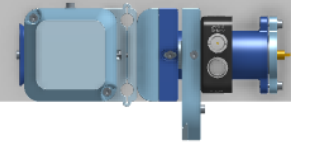
Laser calibration



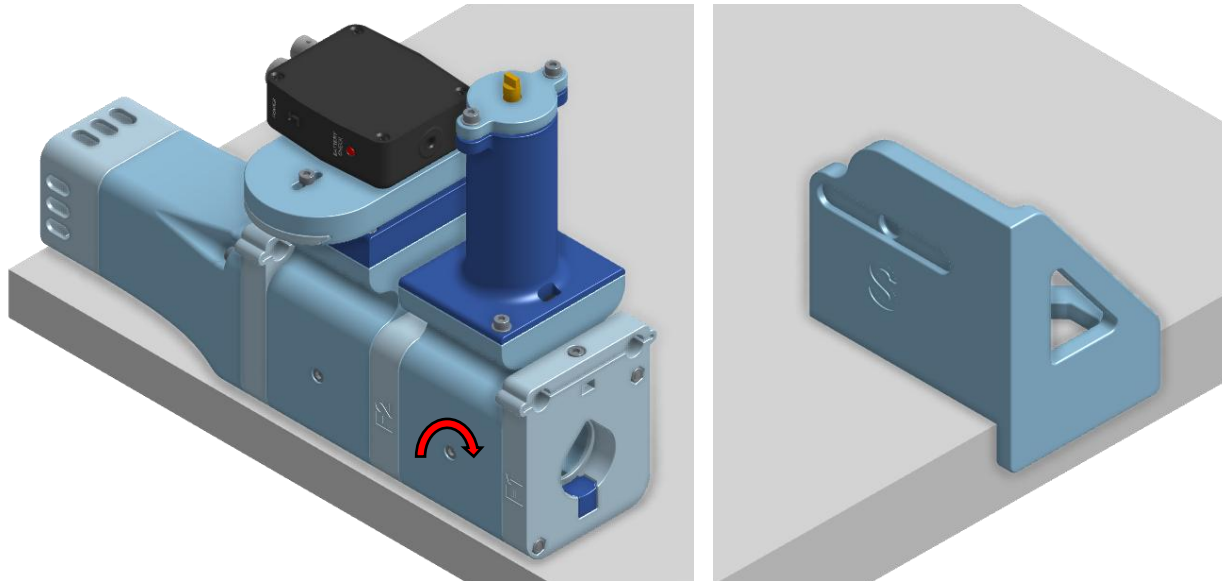
To get the laser as straight as possible, the table calibrator (with an M5 ring) should be placed on the edge of a table. On the other side, the tower assembly (without the bottom cap and semi-transparent mirror) should be placed with its edge lined up with the table.

When this is in place, wiggle the laser sideways, until the laser is roughly aligned with the hole in the table calibrator. After, rotate the end of the laser to create a parallel beam. The beam is parallel when the laser spot is the same size as the inner diameter of the M5 ring.

When the laser is approximately in the right position, glue it down to prevent misalignment.



Laser calibration



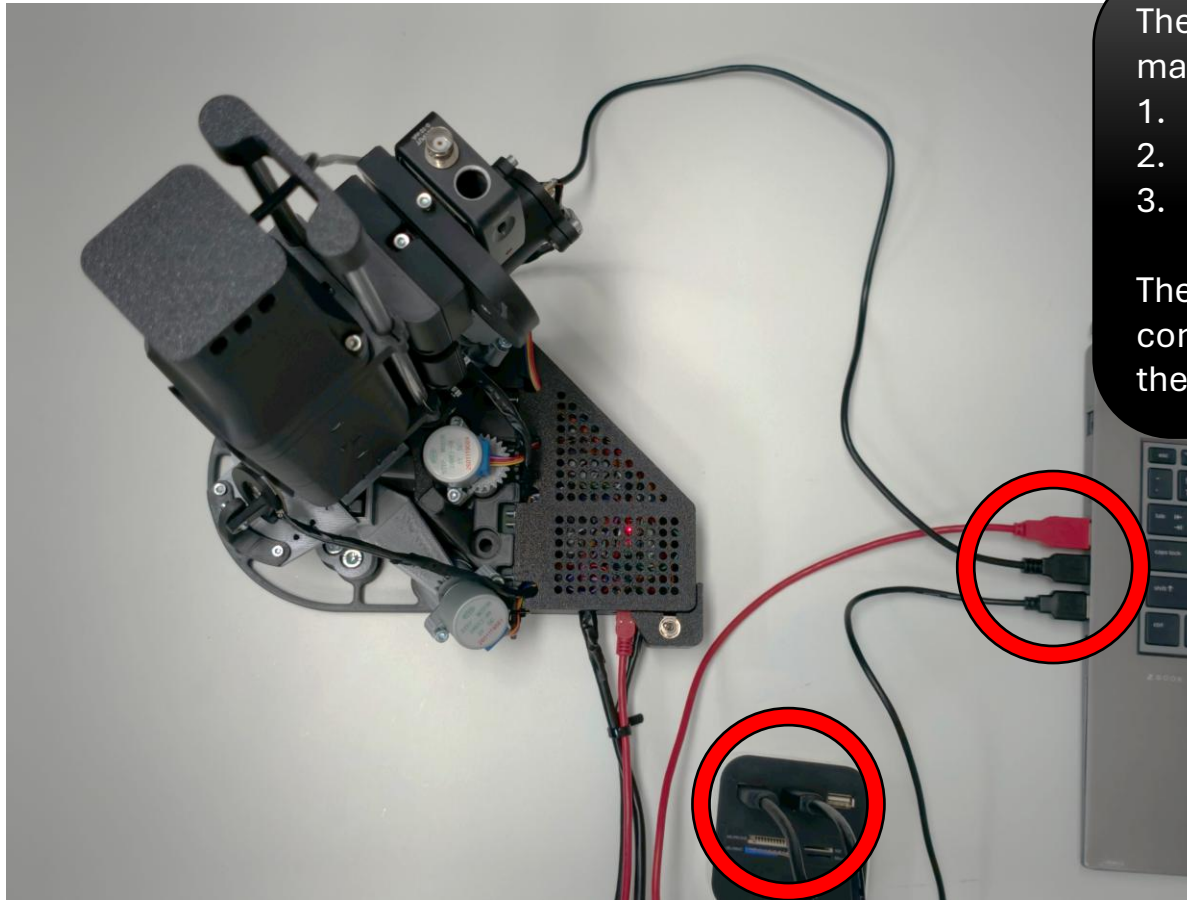
When the laser is fully aligned, the bottom semi-transparent mirror can be calibrated. This is done by removing the F1 and F2 lenses, with its corresponding bolts. The bottom semi-transparent mirror should be mounted back into the system.

When also the two front caps are removed, the tower is aligned with the edge of the table (on the other side). The table calibrator is flipped and placed on the other end.

Now, by rotating the bottom bolt, the beam can be focused onto the calibrator. When the beam is at the same height as the slot (on the calibrator), the beam is exactly reflected at an angle of 90 degrees.



Powering on

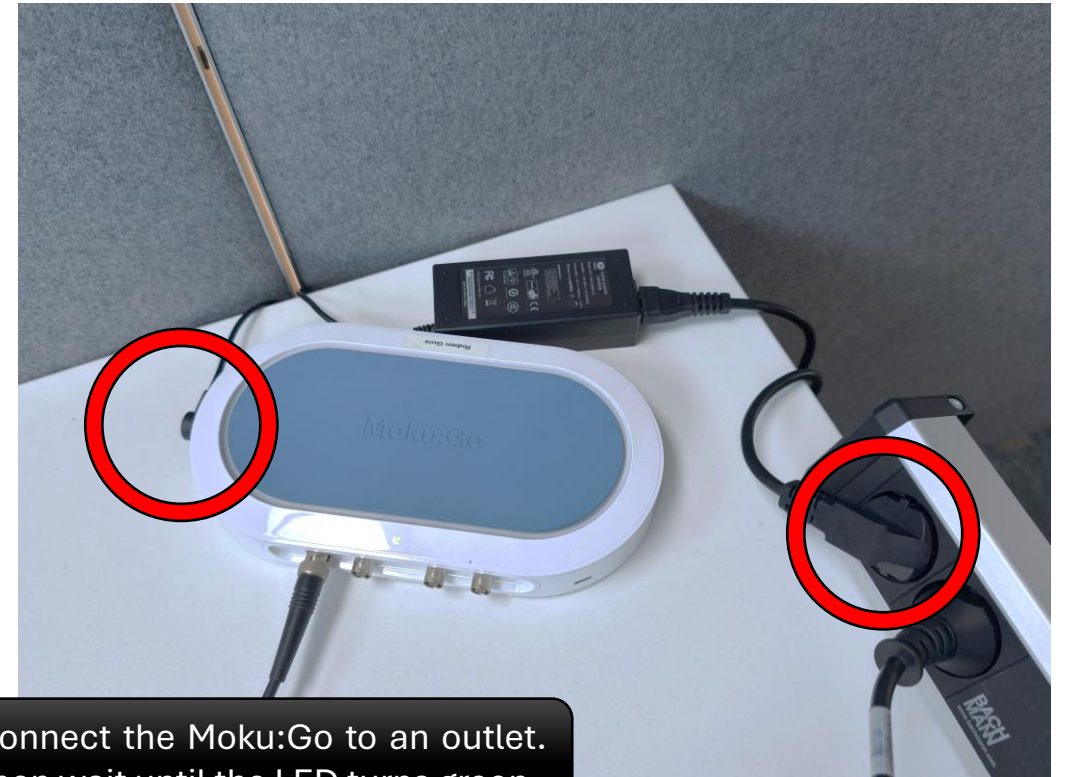
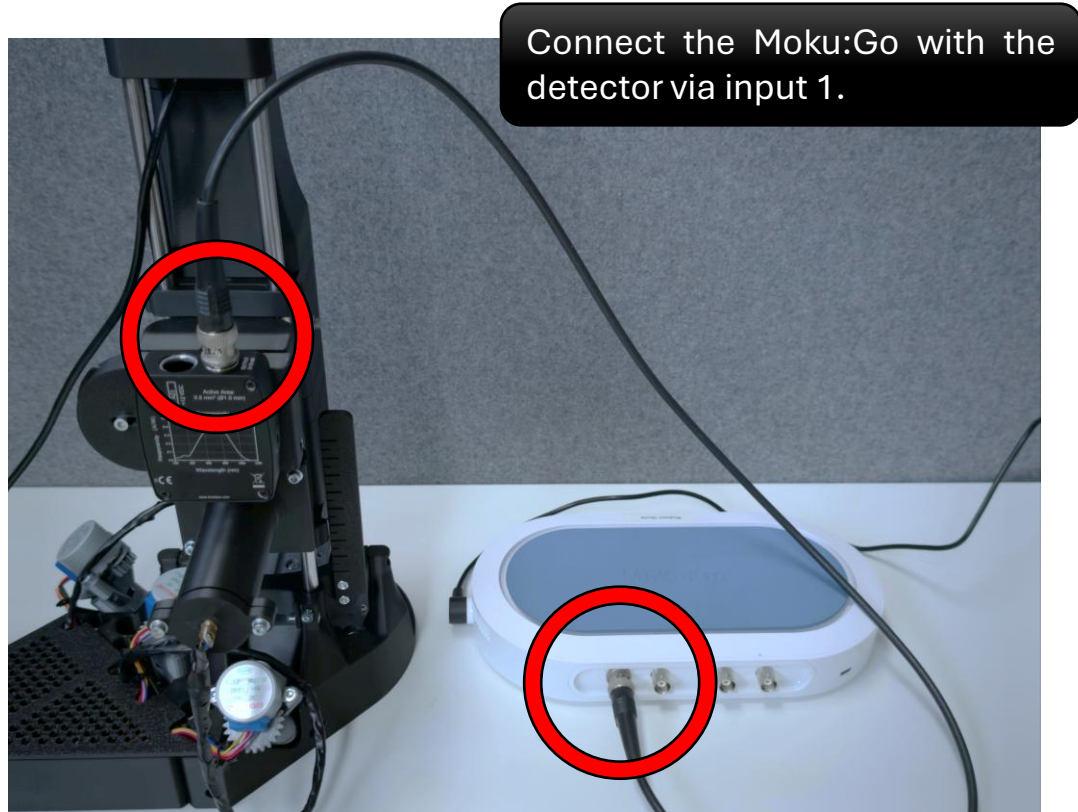


The following USB-connections have to be made with the laptop directly:

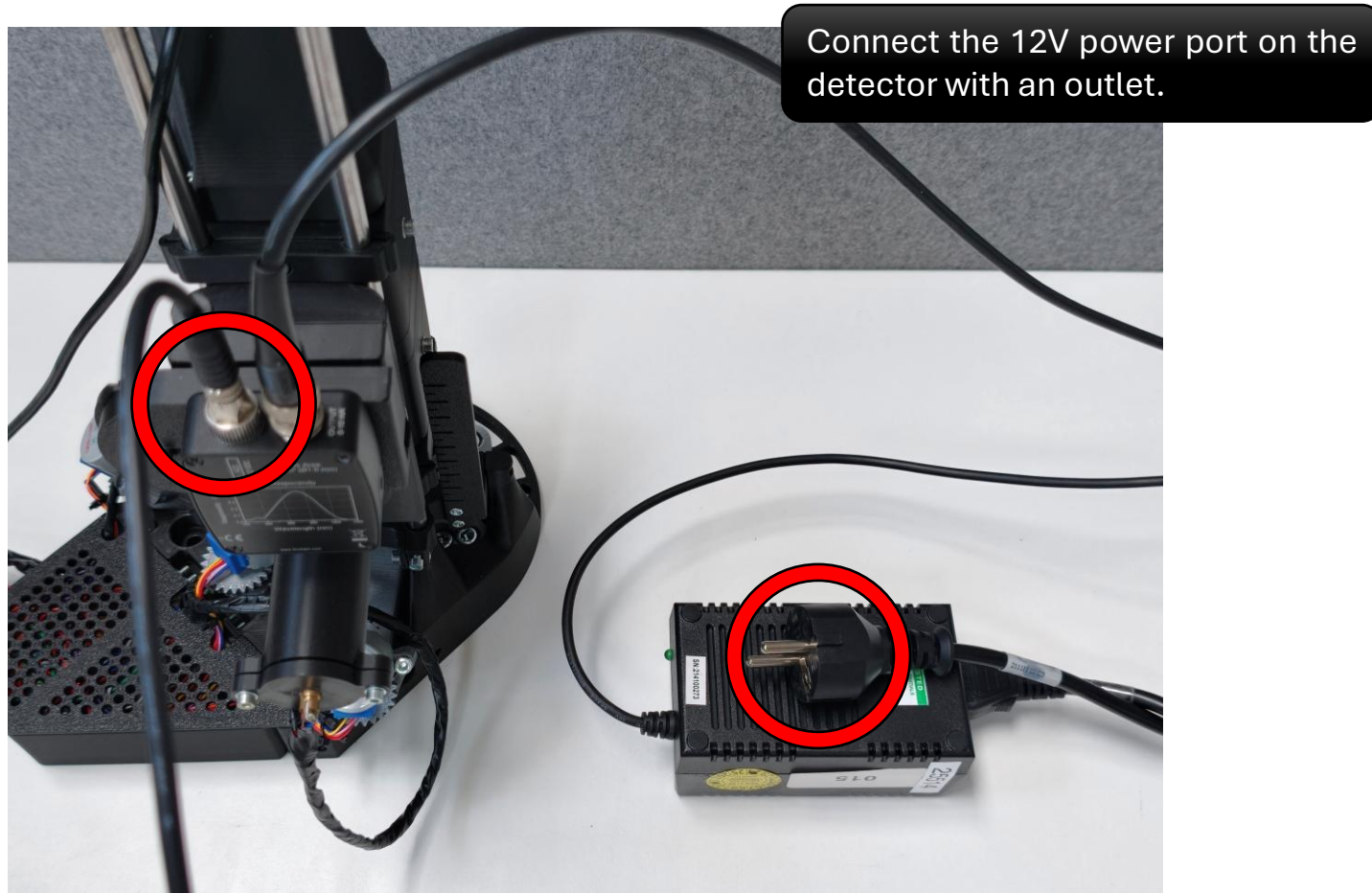
1. Arduino Nano (red)
2. Camera
3. USB-hub

The USB-hub used has two cables connected. One for the laser and the other for the LEDs and motors (see circuit diagram).

Powering on



Powering on



Powering on



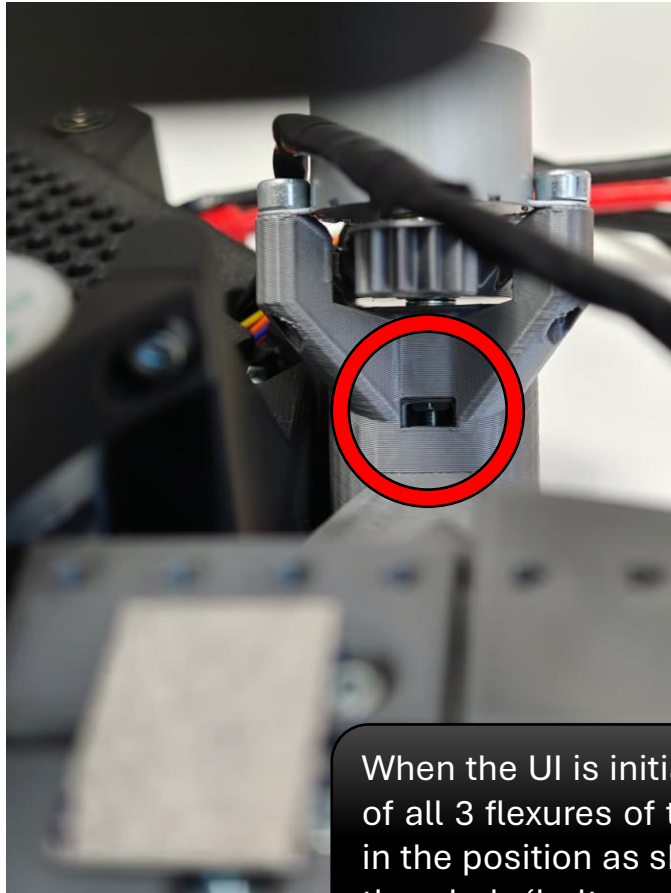
Make sure the power adapter for the 12V power is set to the correct outlet voltage (in this case 230V). After everything is plugged in, turn on the adapter.

After everything is plugged in, do not forget to turn on the detector power as well.

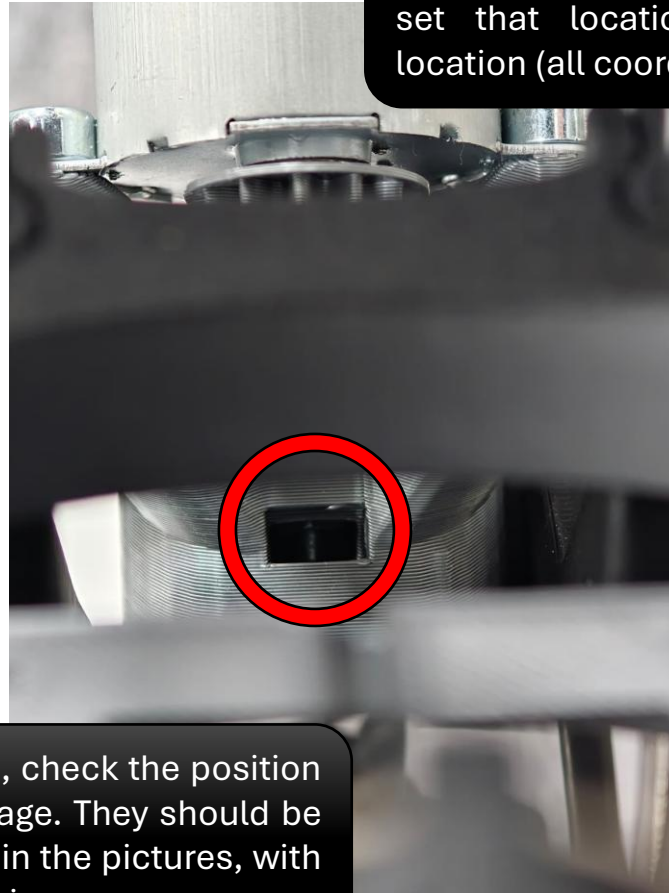


Now that everything is powered on, ui.py can be opened.

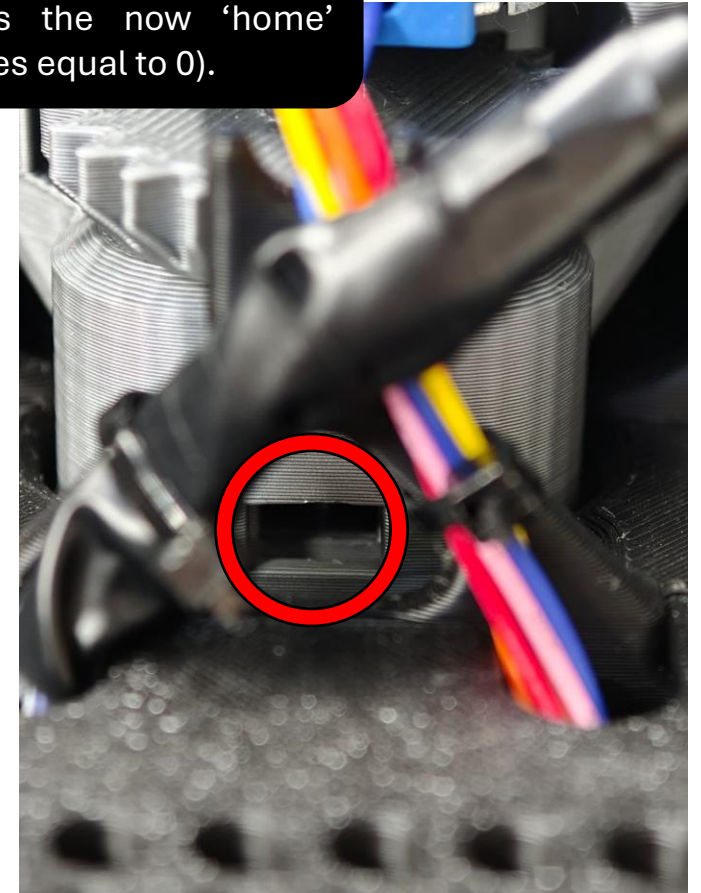
Homing the stage



When the UI is initialized, check the position of all 3 flexures of the stage. They should be in the position as shown in the pictures, with the whole 'bolt-gap' showing.



If the flexures are not in the right location, move the stage using the UI until it is. Then, set that location as the now 'home' location (all coordinates equal to 0).



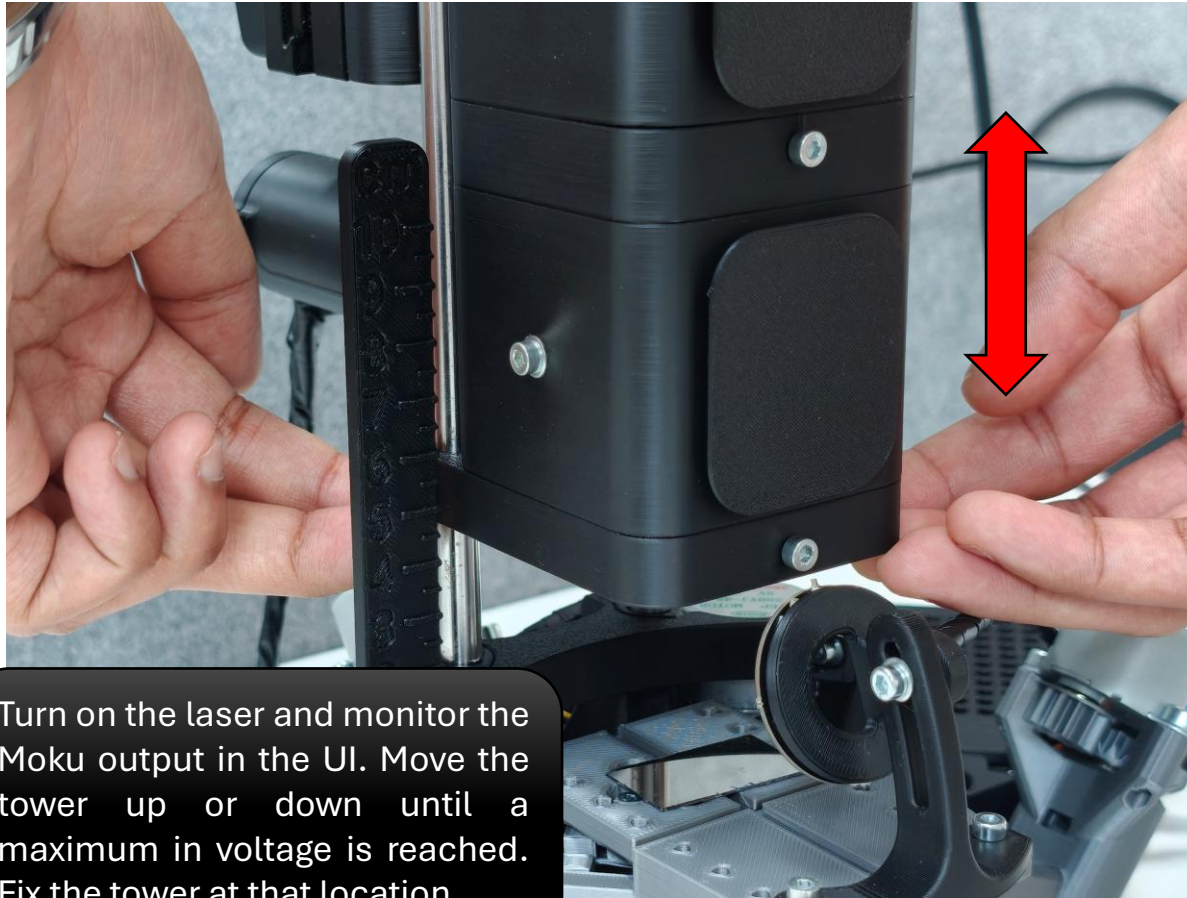
Calibration | Finding I_0

Add an F1 lens of your choice (16-80mm) and place a clean sample/mirror on the stage.



Before the next step, always make sure the top red-light mirror is set to approximately 45 degrees and the sideways slider of the detector is placed in the middle.

Calibration | Finding I_0

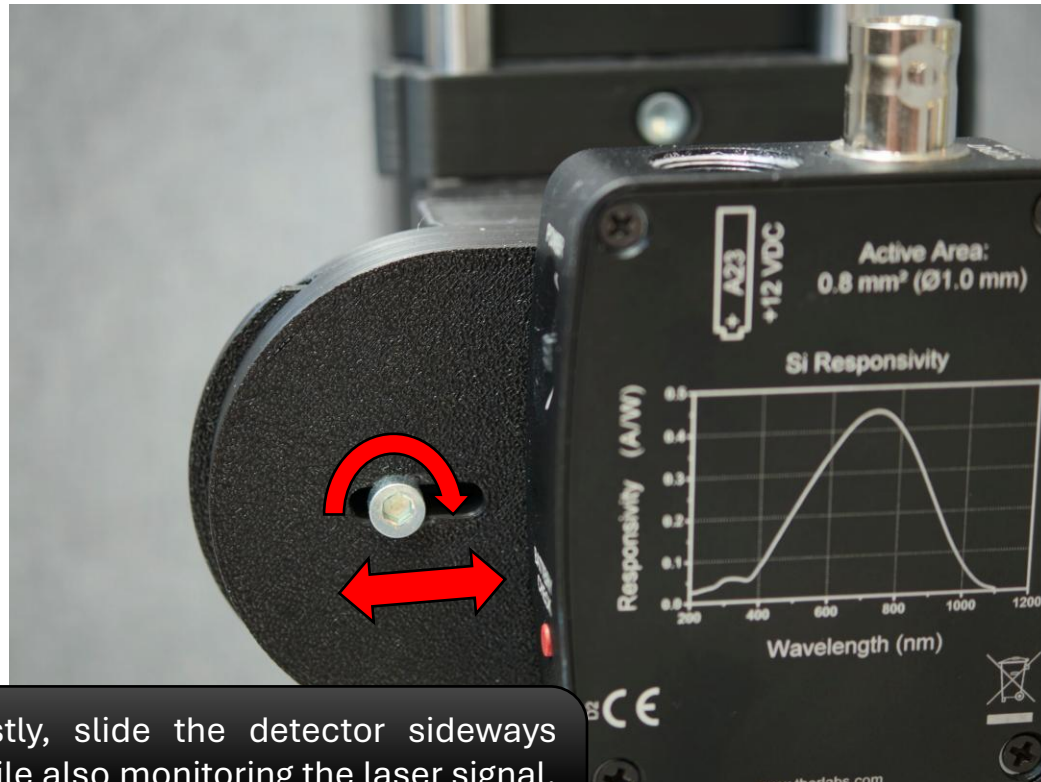


Turn on the laser and monitor the Moku output in the UI. Move the tower up or down until a maximum in voltage is reached. Fix the tower at that location.

Now, carefully rotate the top mirror until the signal of the detector reaches a maximum again. Beware, the alignment of this mirror is very sensitive.



Calibration | Finding I_0



Lastly, slide the detector sideways while also monitoring the laser signal. Make sure to tighten the bolt when the detector is in the right spot.

To check if the maximum intensity has been found, the tower can be moved up and down again (without rotating the mirror or moving the detector). When the maximum voltage is reached, its value should be written down (as well as put into ui.py).

Calibration | Moving to $\frac{1}{2} I_0$

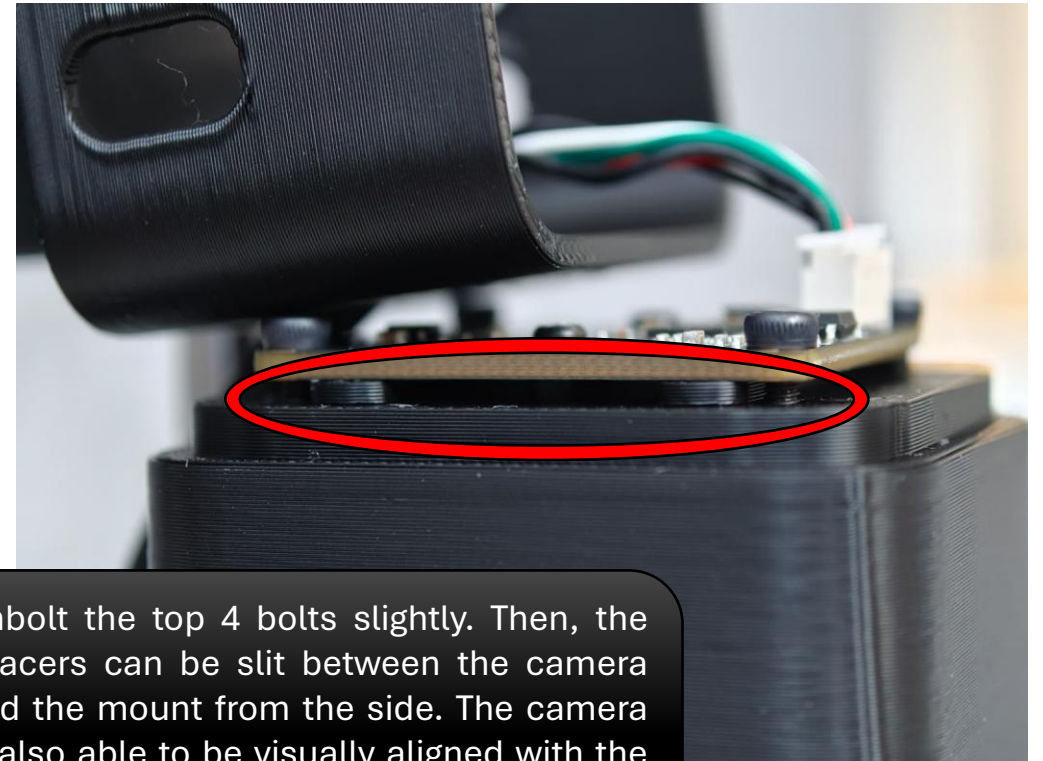


Divide the maximum intensity found in the previous step by 2 and move the tower UPWARDS until the exact $\frac{1}{2} I_0$ voltage is reached. After, secure the tower.

Calibration | Camera calibration

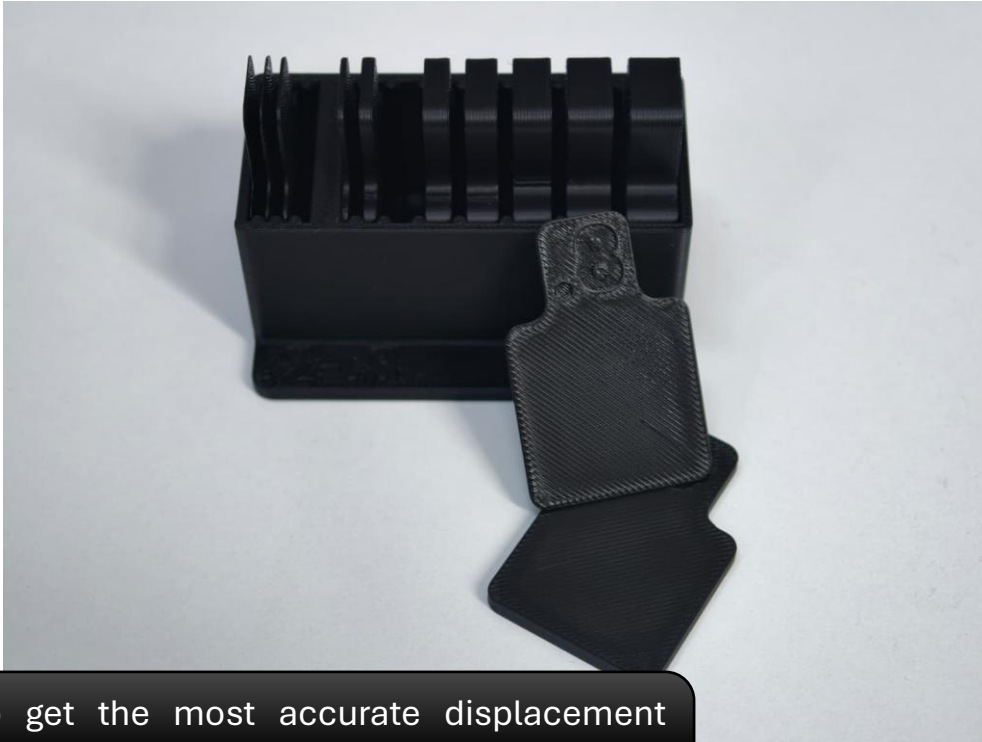


When the signal is at half intensity, the camera should be exactly in focus. Using the camera spacers, the camera can be focused. A 1,2mm spacer was best in our case.

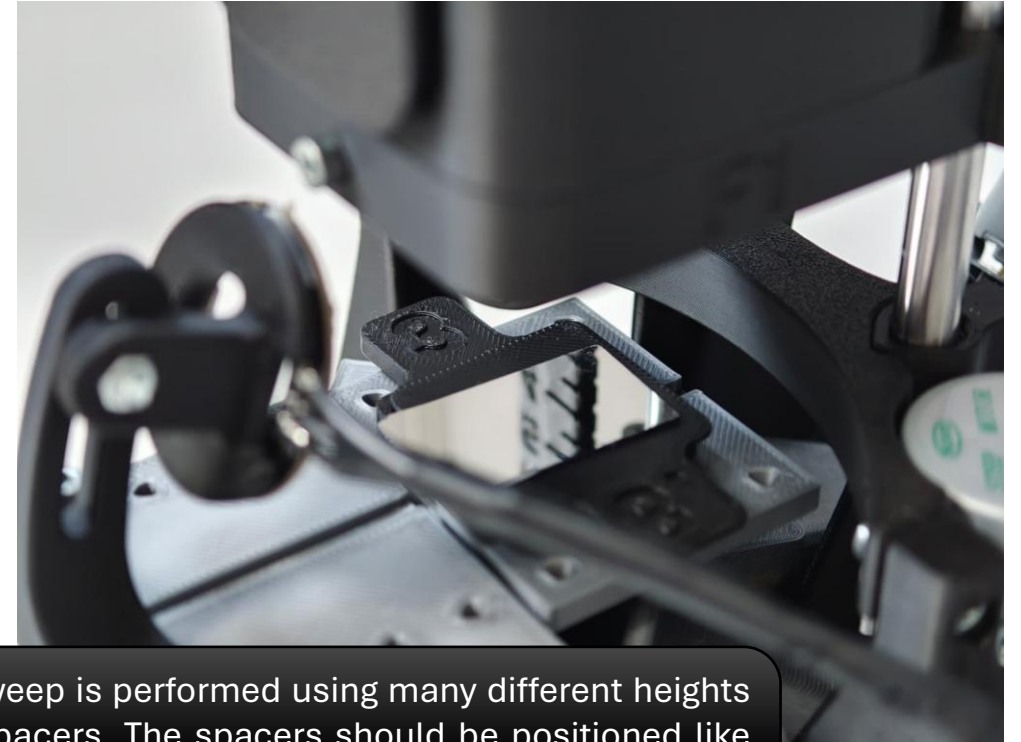


Unbolt the top 4 bolts slightly. Then, the spacers can be slit between the camera and the mount from the side. The camera is also able to be visually aligned with the center of the laser. The camera calibration only needs to be performed once.

Calibration | Performing a sweep



To get the most accurate displacement measurements, the system gets calibrated using spacers with known heights.



A sweep is performed using many different heights of spacers. The spacers should be positioned like in the image above, without overlapping numbers (more accuracy due to printing defects).

Calibration | Performing a sweep

	A	B
1	-4	1,16
2	-3,6	1,45
3	-3,2	1,58
4	-2,8	1,81
5	-2,4	2,11
6	-2	2,38
7	-1,6	2,78
8	-1,2	3,07
9	-0,8	3,67
10	-0,4	3,96
11	0	4,36
12	0,4	5,27
13	0,8	5,68
14	1,2	6,6
15	1,6	7,6
16	2	8,04
17	2,4	8
18	2,8	7,52
19	3,2	6,44
20	3,6	5,74
21	4	5,46

To start of the sweep, position the 4mm spacer and a mirror on top on the stage. Then, move the tower to find I_0 (and align the top mirror and detector if needed). After, move the tower to $\frac{1}{2} I_0$.

Now, open an empty Excel document (or csv editor) and create two columns. The first column being the spacer height value, and the second column being the measured output voltage of the detector.

Set the 4mm spacer equal to a height of 0 and add the corresponding voltage value to the table. Now, switch out the spacers in steps of 0,4 or 0,2mm (depending on the size of F1) and measure the voltage at each height. This will create a result as shown in the picture on the left. This is a sweep with steps of 0,4mm with an F1 lens of 40mm.

The Excel document will be used in the ui.py to calibrate the displacement measurements. Only one sweep per F1 lens needs to be performed.